Jung et al. 2018, ApJ, 864, 103

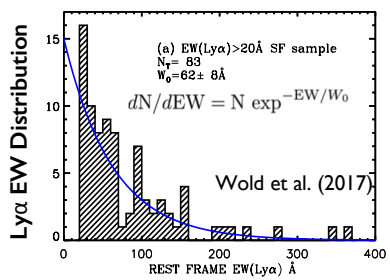
Texas Spectroscopic Search for Lyα Emission at the End of Reionization Constraining the Lyα Equivalent Width Distribution at z > 6

Intae Jung[1], Steven Finkelstein[1], Rachael Livermore[1,2,3], Mark Dickinson[4], Rebecca Larson[1], Casey Papovich[5], Mimi Song[6], Vithal Tilvi[7], and Isak Wold[1]

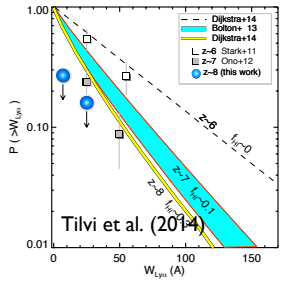
[1]The University of Texas at Austin, [2]The University of Melbourne, [3]ASTRO 3D, [4]National Optical Astronomy Observatory, [5]Texas A&M University, [6]NASA Goddard Space Flight Center, [7]Arizona State University

[1] Topic: Lyα Emission as a Probe of Cosmic Reionization

- Neutral hydrogens in the IGM affect the observed features of Lyα emission



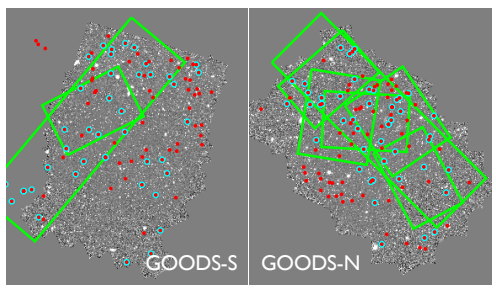
Lyα galaxy EW distribution with the best fit exponential, well characterized by e-folding scale, w0.



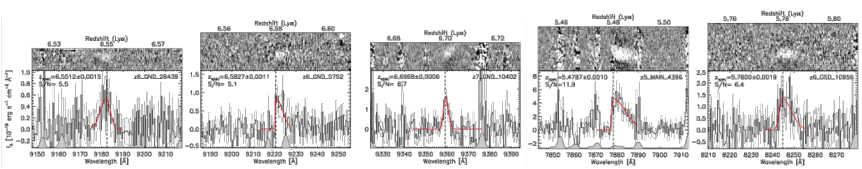
Observed Lyα EW distribution with the model predictions

[2] Spectroscopic Observation for z = 5.5 – 8.2 galaxies

The DEIMOS (optical) and MOSFIRE (NIR) spectrographs on the Keck telescope



The DEIMOS & MOSFIRE slitmask configurations



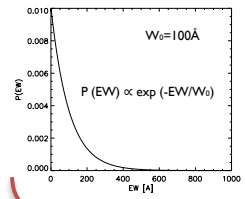
Five detected Lyα emitters from the DEIMOS data

[3] Measuring the Lyα Equivalent Width Distribution

★ Simulating the expected number of detections accounting for incompleteness (e.g.,, spectroscopic depth, sky emission, wavelength coverage, P(z) distribution)

- We simulate mock emission lines for our observed galaxies in a Monte-Carlo fashion.

i) Lyα strength from P(EW)



ii) Line location from P(z)

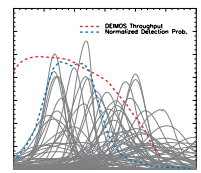
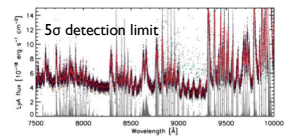
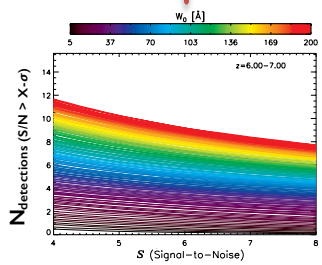


Photo-z PDF

iii) Determine S/N levels



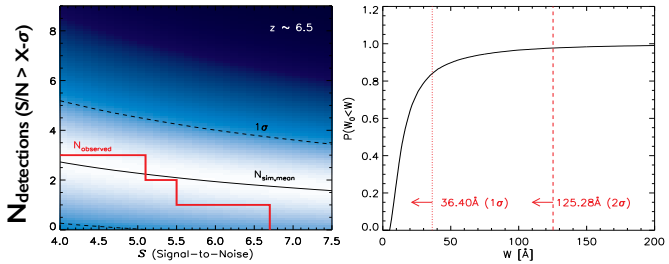


- From the simulations, we calculate the posterior distribution of the expected number of detections as a function of S/N for e-folding scales of W_0=5-200Å.

← The mean expected number of detections as a function of S/N for a range of EW distributions for z ~ 6.5.

[4] Lyα Equivalent Width Distribution at z ~ 6.5

From the MCMC sampling using the simulated expected number of detections with the detections from observations, we calculate the posterior distribution of the Lyα EW e-folding scale (W_0).



Constrained probability of EW distribution with observed Lyα detections at z ~ 6.5.